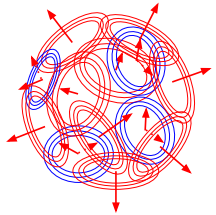
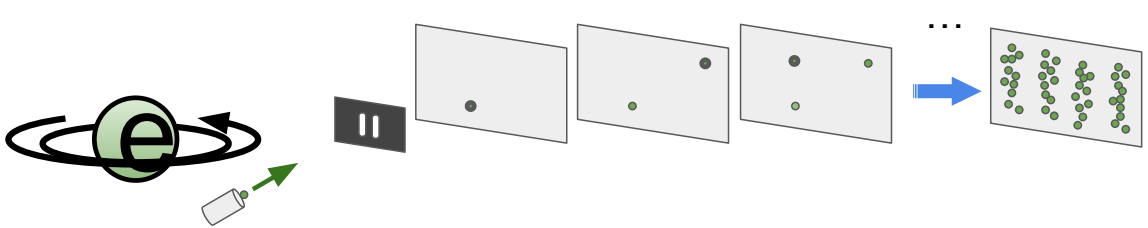
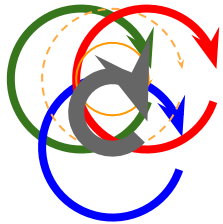
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.5: Electron Double-Slit Experiment





1. De Broglie’s Matter Waves

In 1905, Albert Einstein proposed the photon hypothesis, suggesting that light, which was previously thought to be a wave (electromagnetic wave), also possesses particle-like properties.

Conversely, in 1924, Louis de Broglie proposed that if light has particle properties, then particles with mass, such as electrons, should also possess wave-like properties (wavelength).

This wave-like nature of particles is called a "matter wave" or a "de Broglie wave".

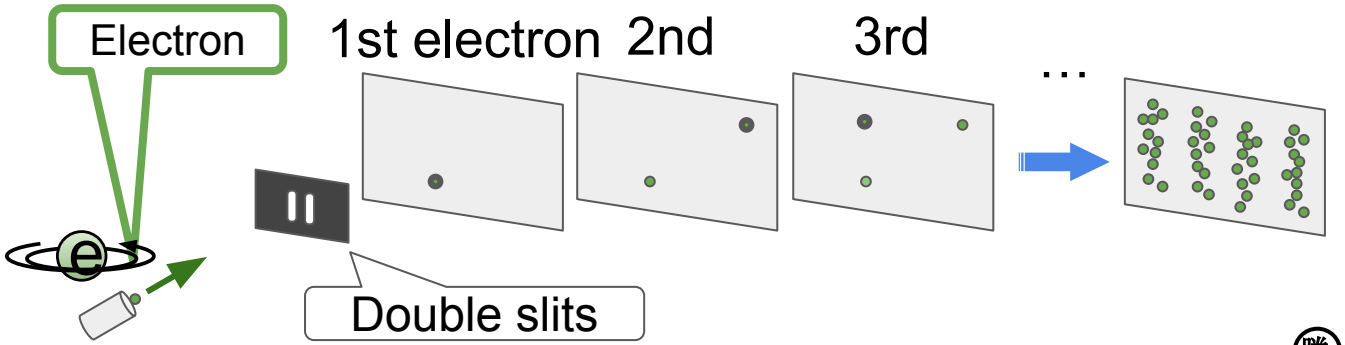
Matter waves were later proven through experiments such as electron diffraction.

1. De Broglie's Matter Waves

This concept led to the further development of quantum mechanics, including Schrödinger's wave function.

An experiment can be conducted by replacing the photons in the double-slit experiment described in Chapter 4 with electrons.

When electrons are emitted toward a double slit one by one, a similar interference pattern appears.



1. De Broglie’s Matter Waves

The mysterious phenomenon where a single electron passes through two slits simultaneously and behaves as if it were interfering with itself was confirmed by Akira Tonomura and others in their 1989 "Electron Double-Slit Experiment" .

This phenomenon is caused by de Broglie's matter waves.

The wavelength of a matter wave is expressed as ”$\lambda=h/p$”.

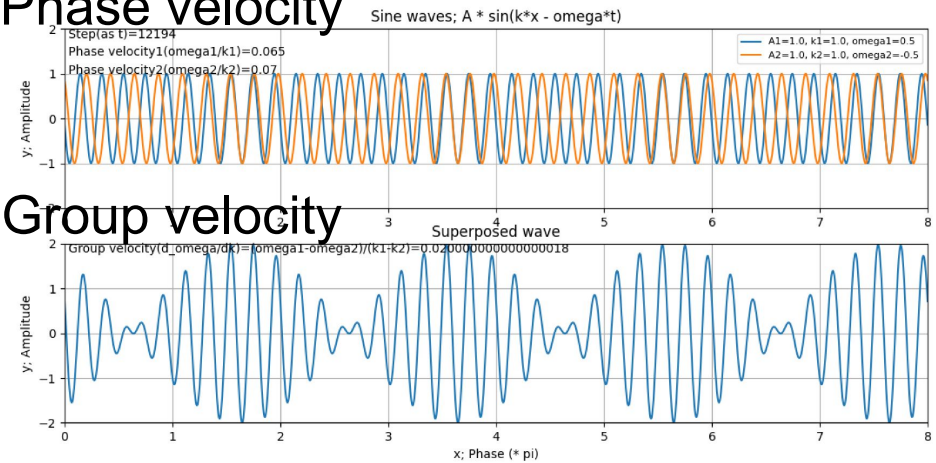
By combining this with the Planck-de Broglie relations ($E=\hbar\omega$, $p=\hbar k$), two different wave velocities appear for a particle with mass: "phase velocity" and "group velocity".

1. De Broglie’s Matter Waves

Phase velocity is the speed at which the individual peaks (phases) of the wave progress.

Group velocity is the speed at which the "beat" (wave packet or pulse) formed by the superposition of several waves progresses.

Phase velocity



Group velocity

1. De Broglie’s Matter Waves

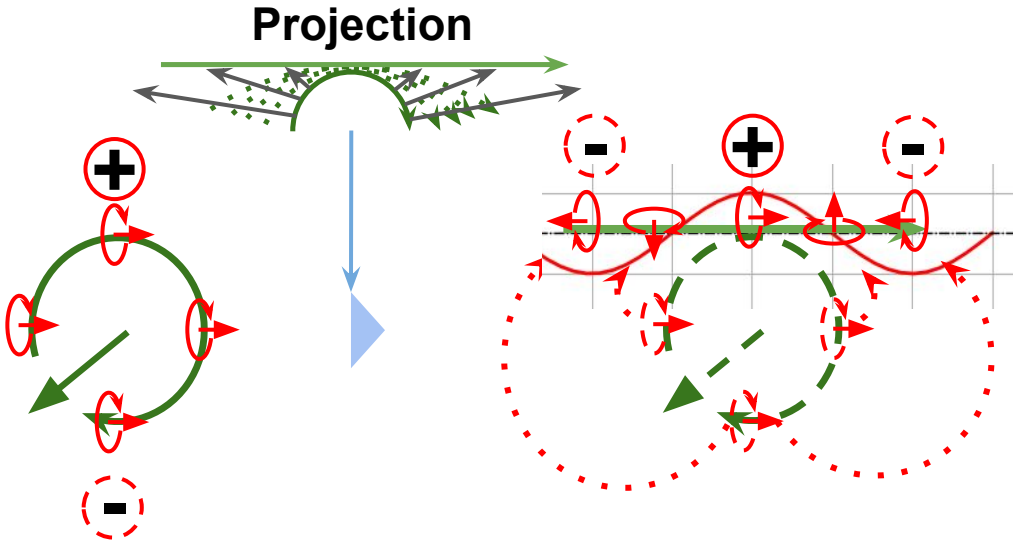
In quantum mechanics, the "clump of energy or information where a particle actually exists" moves at this group velocity, and the classical velocity of a quantum coincides with the group velocity.

In quantum mechanics, particles are viewed as a superposition of probability waves.

Just like photons, the probability waves of electrons pass through both slits simultaneously and interfere, causing an interference pattern to appear on the observation screen.

2. Wave Nature of the Electron Model

In R-space Theory, the mechanism by which wave properties arise from the rotation vector (angular momentum) of a photon is due to the alternating orientation of the rotation vector when its phase in R-space is projected onto spacetime, as explained in Chapter 4.

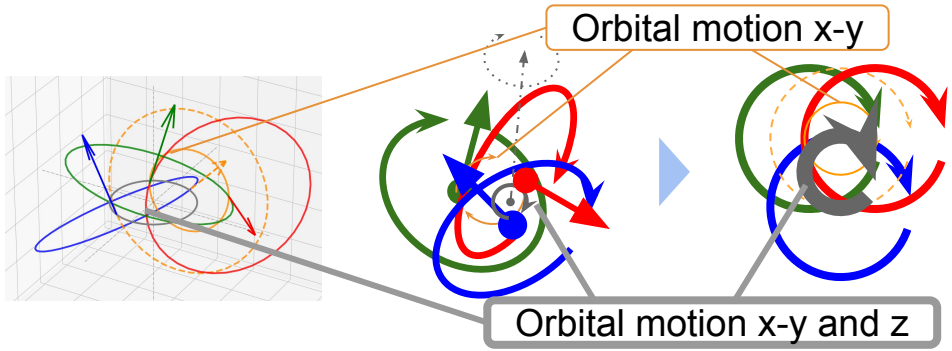


2. Wave Nature of the Electron Model

The case for electrons is similar but slightly different because electrons involve a "double rotation" of the rotation vector.

To illustrate the double rotation of the electron's rotation vector clearly, it is diagrammed in a plane.

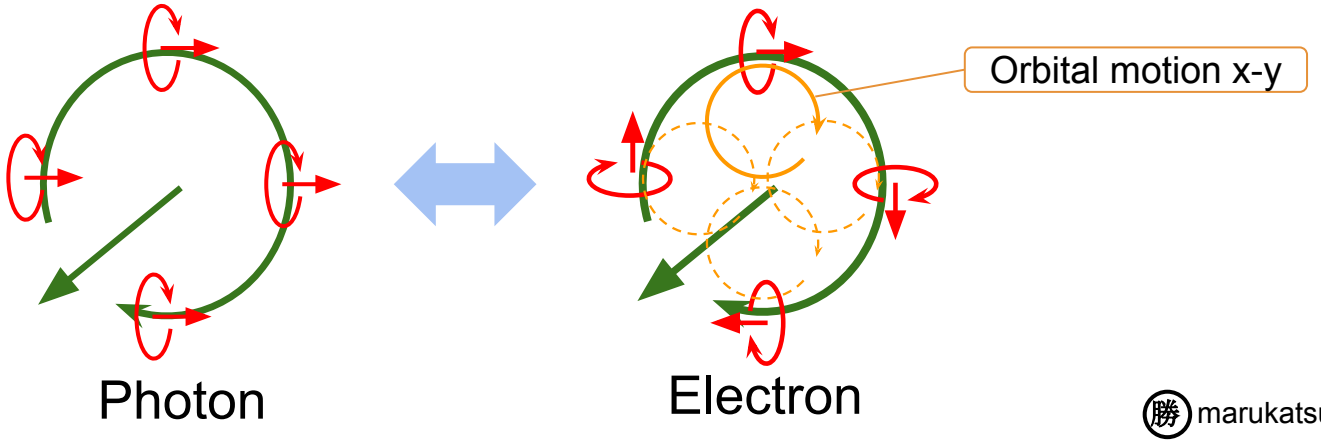
It can be seen that three rotation vectors (x, y, z) revolve around each other like a binary star system, performing a double rotation.



2. Wave Nature of the Electron Model

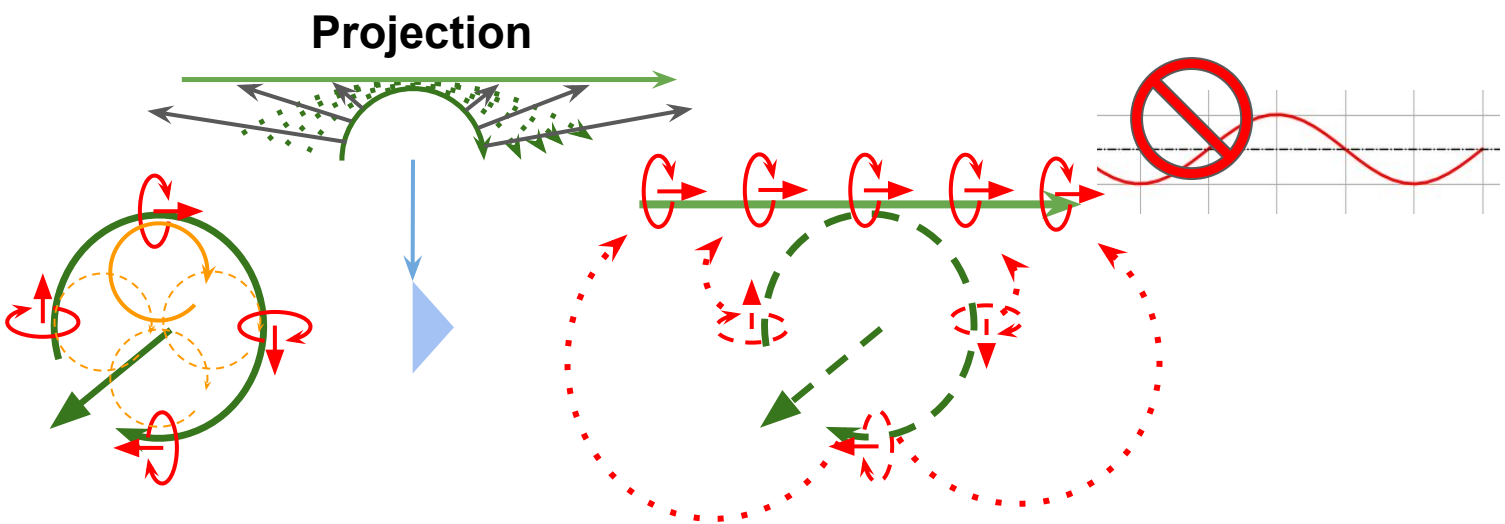
Next, focusing on rotation vectors x (red) and y (green), as in the explanation of the photon: in the case of a photon, the direction of rotation vector x is the same at any phase.

However, in the case of an electron, rotation vectors x and y revolve around each other (yellow), so their orientation changes depending on the phase.



2. Wave Nature of the Electron Model

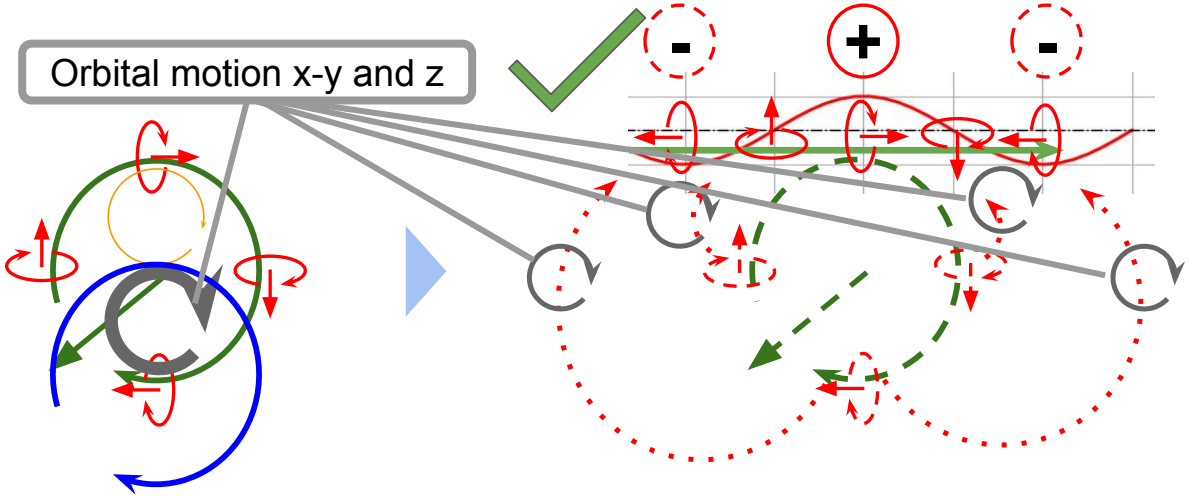
When the phase circle in R-space is expanded and projected onto spacetime, the orientation of rotation vector x points in the same direction in space, so wave properties do not appear.



2. Wave Nature of the Electron Model

However, rotation vector x (red) is further rotating due to its orbital motion (grey) with rotation vector z (blue).

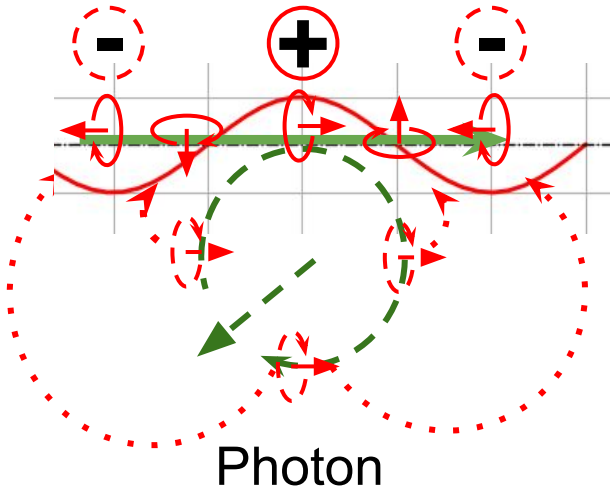
Therefore, the orientation of rotation vector x rotates according to the phase of this orbital motion (grey), pointing right, up, left, and down.

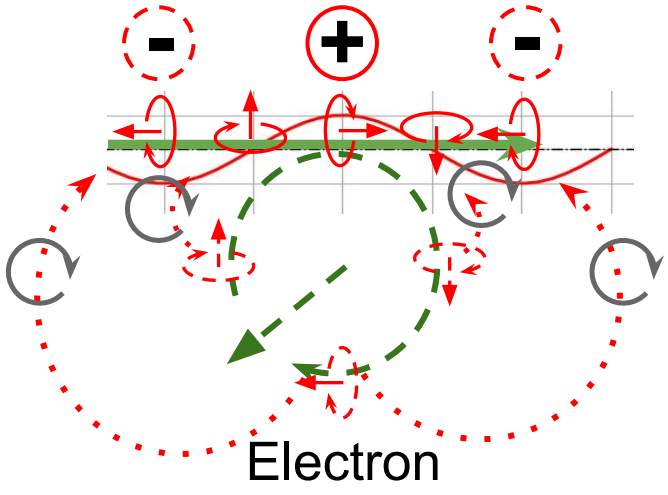


2. Wave Nature of the Electron Model

If right is positive and left is negative, it alternates between positive and negative.

In this way, wave properties are generated from the rotation vector even in the case of an electron.



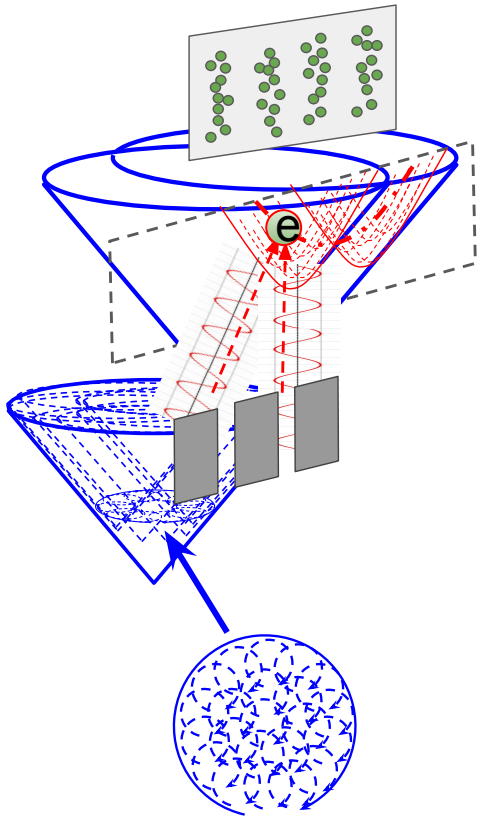


2. Wave Nature of the Electron Model

The electron's rotation vector is a collection (superposition) of angular momentum vortices, just like a photon, and the mechanism behind the mysterious phenomenon in the double-slit experiment is the same as that of the photon explained in Chapter 4.

When passing through the double slit, the internal phase of the electron splits into two and causes interference;

however, because the electron is projected into spacetime probabilistically according to the density of the superimposed phases, it behaves like a single particle when observed on the screen.



2. Wave Nature of the Electron Model

Note that, as will be explained in the following chapters, photon waves are projected in proportion to distance in Minkowski space, whereas electron waves are projected in proportion to the "square of the distance" due to the influence of the double rotation.

The idea that an electron is a rotated photon has already been proposed in the "Williamson & van der Mark (WvdM) electron model", so R-space Theory is not the first to suggest this.

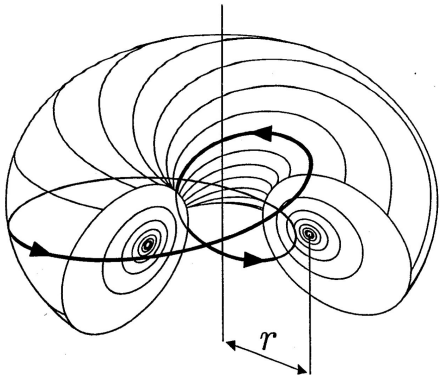
Additionally, the idea that Minkowski space is a secondary structure derived from a projection—which in R-space Theory is the projection of phase and rotation in R-aether—has already been proposed in "Twistor Theory".

3. Williamson & van der Mark (WvdM) Electron Model

The Williamson–van der Mark electron model was published in 1997 by J.G. Williamson and M.B. van der Mark in their paper "Is the electron a photon with toroidal topology?".

In this model, an electron is defined as a "single photon self-confined in a toroidal (donut-shaped) closed path".

It posits that an electron is formed when a circularly polarized photon (electromagnetic wave) forms a twisted loop, resulting in a toroidal structure with a double loop.



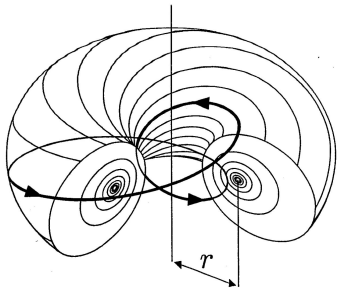
WvdM Electron Model

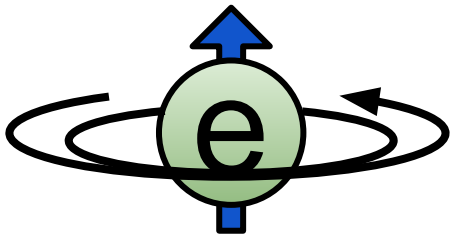
References:"Is the electron a photon with toroidal topology?" J.G. Williamson and M.B. van der Mark.

3. Williamson & van der Mark (WvdM) Electron Model

Because the electron is a rotating photon and the photon continues to circulate at the speed of light internally, this is consistent with "Zitterbewegung (jittery motion)" derived from the Dirac equation and the interpretation of matter waves.

Furthermore, the property where the phase returns to its original state after a “4 * π” rotation due to the photon's polarization on the torus and the double-loop structure represents "1/2 rotational symmetry", which explains the electron's "1/2 spin".

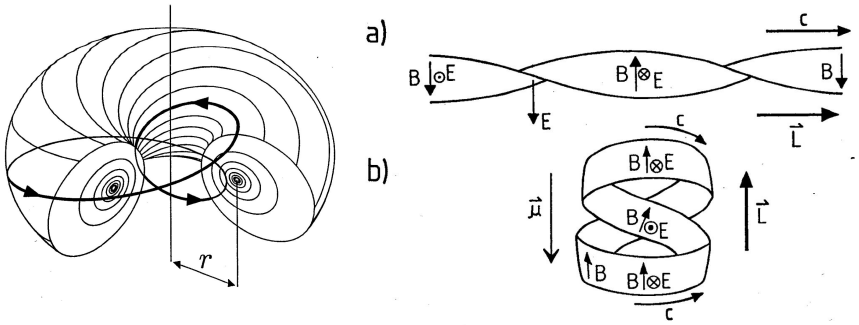


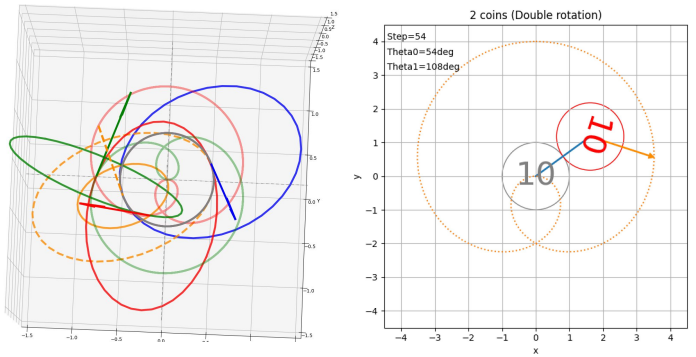


3. Williamson & van der Mark (WvdM) Electron Model

Additionally, the toroidal confinement of the photon's electromagnetic field explains the electron's charge and magnetic moment.

The WvdM model is similar to the electron model in R-space Theory in that it considers the electron to be a rotated photon and explains the 1/2-Fold Rotational symmetry of electron spin through a toroidal structure.





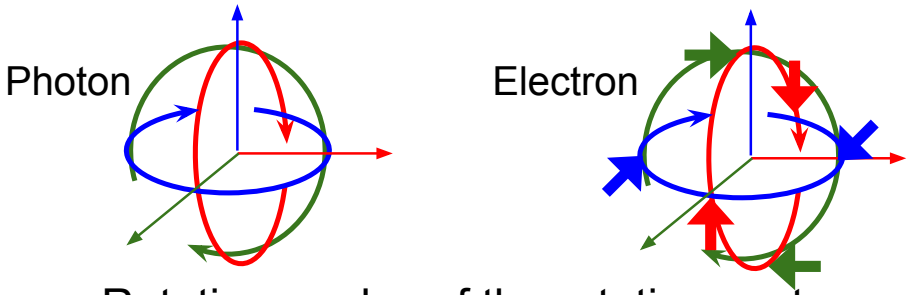
R-space Theory Electron Model

References:"Is the electron a photon with toroidal topology?"
J.G. Williamson and M.B. van der Mark.

3. Williamson & van der Mark (WvdM) Electron Model

The WvdM model introduces non-linear electromagnetic fields for the photon to become self-confined; similarly, R-space Theory posits that the rotation modes of the rotation vectors for photons and electrons change according to the magnitude of rotational energy.

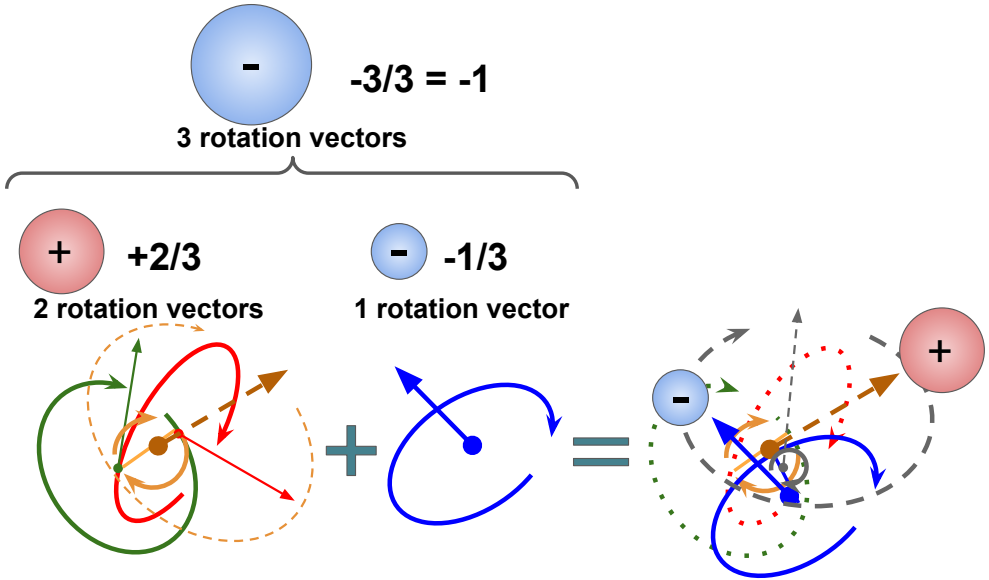
However, they differ in that in R-space Theory, the three rotation vectors in both photons and electrons rotate each other, meaning both are inherently in a state of self-confinement.



Rotation modes of the rotation vectors

3. Williamson & van der Mark (WvdM) Electron Model

Another difference is that in R-space Theory, as explained in Chapters 2 through 4, charge, magnetic moment, and electromagnetic waves are all manifestations of the rotation vector (angular momentum).



4. Twistor Theory

Twistor Theory is a theory proposed by "Roger Penrose" in 1967 that attempts to unify general relativity and quantum mechanics.

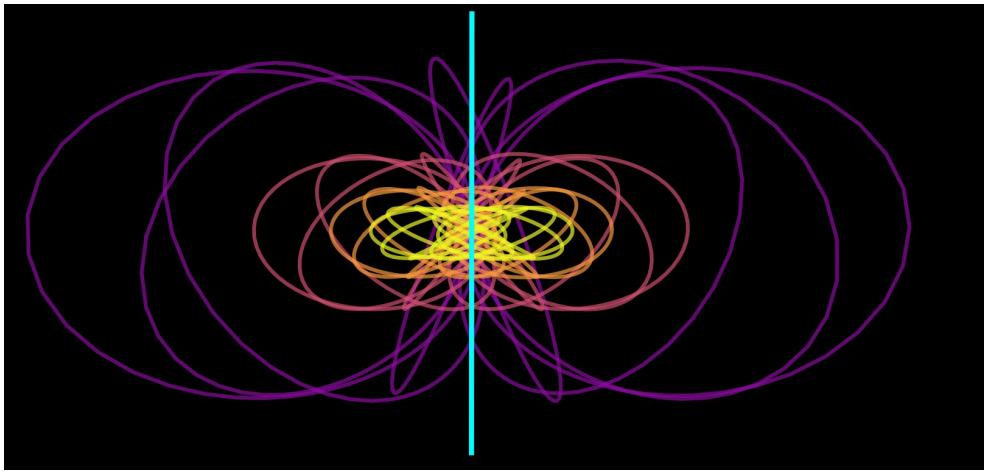
It proposes that the fundamental root of our world is not Minkowski space, but "Twistor space", which is a complex spinor space, and that Minkowski space is merely a secondary structure emerging from it.

R-space Theory shares the view that Minkowski space is secondary, as it considers Minkowski space to be a projection of R-space, which consists of rotation and phase in R-aether.

4. Twistor Theory

The visual representation of a twistor by Ivor Robinson (Robinson Congruence) is depicted as a set of twisted lines in 3D space (a collection of non-intersecting straight lines).

This represents how a point in 4D Twistor space appears in our 3D spacetime.



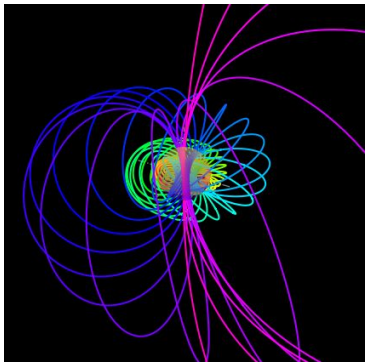
Robinson Congruence

4. Twistor Theory

Geometrically, it has essentially the same structure as a stereographic projection of a "Hopf fibration"

and consists of a collection of lines spiraling around the surface of a torus (donut shape).

These twisted donut-shaped geometric forms, such as toroids and tori, are concepts common to the WvdM model, Twistor Theory, and R-space Theory.



Hopf fibration

5. Comparison of the Three Theories

The following table compares the Williamson & van der Mark electron model (WvdM model), Twistor Theory, and R-space Theory.

Feature	WvdM Model	Twistor Theory	R-space Theory
Spacetime	Assumes normal Minkowski space.	✅ Minkowski space is derived from Twistor space.	✅ Minkowski space is a projection of rotation and phase in R-aether.
Electron Picture	✅ Photon rotates in a closed toroidal shape (electron has internal structure).	Electron is expressed as a spinor/Dirac field; internal structure is not directly depicted.	✅ Rotation vectors (angular momentum) revolve in a double rotation (electron has internal structure).

5. Comparison of the Three Theories

Feature	WvdM Model	Twistor Theory	R-space Theory
Origin of Spin	✅ Geometrically explains spin 1/2 from toroidal circulation/rotation.	Explains spin from spinor representation (SL(2,C)) and the 4-component structure of twistors.	✅ Geometrically explains spin 1/2 from the double rotation of rotation vectors.
Origin of Charge	Arises from the topology/circulation of electromagnetic waves (light).	✅ In Super-twistor, a "charge-like" quantity appears as U(1) internal symmetry.	✅ The rotation vector (angular momentum) itself is the charge.

5. Comparison of the Three Theories

Feature	WvdM Model	Twistor Theory	R-space Theory
Origin of Light speed constancy	Defined as the motion of the photon inside the electron.	Twistor space is a "mathematical structure that preserves the light-cone structure invariant to the speed of light."	Phase restriction during interaction between a photon (spin 1) and an electron (spin 1/2) (due to revolution and precession of rotation vectors).
Wavefunction Collapse	The concept of wavefunction collapse does not exist.	Objective Reduction (OR) hypothesis due to gravity.	Phase superposition of microscopic vortices (angular momentum) inside the quantum acts like a pilot wave.

5. Comparison of the Three Theories

The "U(1) symmetry" shown by Super-twistor is mathematically a "first-order unitary group," a set of complex numbers “e^iθ” with an absolute value of 1. On the complex plane, this is like a "circle of radius 1" centered at the origin, which can be considered identical to the rotation vector in R-space.

From this comparison, it can be seen that R-space Theory is a theory that adds internal electron structure to Twistor Theory.

However, interpretations regarding the collapse (reduction) of the wavefunction differ.

References : Annales de la Fondation Louis de Broglie, Volume 22 no 2, 1997, 133: "Is the electron a photon with toroidal topology?" J.G. Williamson and M.B. van der Mark.

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This is all for my presentation.

Thanks for watching!

